

Rahul Rawat

MASTER THESIS STUDENT

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PROFILE

Data Science and Analytics Master student at SRH Heidelberg based in Mannheim, with hands on experience building Retrieval Augmented Generation systems and regulator defensible machine learning tools. I have shipped a modular RAG orchestrator on Llama 3.1 with LangChain routing and ChromaDB retrieval, and I documented a full EU AI Act and GDPR compliance case on the CreditIQ credit scoring project, which maps directly onto the AI powered Compliance Assistant thesis brief. I combine that with a six month industry collaboration on time series forecasting and an IEEE style Bachelor Thesis, so I can carry a six month thesis from literature review through evaluation to a working prototype on my own.

EXPERIENCE

eRay GmbH

Oct 2025 to Mar 2026

Data Scientist

- Built an end to end recursive time series pipeline forecasting chlorophyll a, turbidity, pH, and dissolved oxygen for a German lake, delivered as a six month collaboration with SRH University Heidelberg.
- Benchmarked six models head to head, Ridge, Gradient Boosting, LightGBM, XGBoost, CatBoost, and Prophet, and used CatBoost multi quantile regression to produce asymmetric 80 percent prediction intervals.
- Enforced strict anti leakage rules across the pipeline, surfacing the honest finding that pH and dissolved oxygen are physically predictable while chlorophyll a and turbidity are not without live optical sensors.
- Reconstructed missing winter readings with MICE imputation and engineered a synthetic winter decay forecast canvas so tree based models stopped flatlining during recursive prediction.
- Wrapped the whole pipeline in an orchestrator with gate checks and velocity and ecological bounds that halts on failed imputation rather than letting bad data cascade downstream.

EDUCATION

M.Sc. Data Science and Analytics

Apr 2025 to Present

SRH University of Applied Sciences Heidelberg

Bachelor of Technology in Computer Science

2019 to 2023

GL Bajaj Institute of Technology and Management

PERSONAL PROJECTS

Hybrid RAG Orchestrator

Python

LangChain

Llama 3.1 8b on Groq

ChromaDB

HuggingFace MiniLM L6 v2

Streamlit

- Designed a modular Retrieval Augmented Generation system using Llama 3.1 8b via Groq for high speed inference and LangChain for orchestration across multiple execution paths.
- Engineered a custom decision making router that dynamically classifies user intent into three execution paths, local knowledge retrieval on PDF and vector stores, external web search through DuckDuckGo, and direct conversational logic.
- Implemented ChromaDB with local persistence for document storage, using HuggingFace MiniLM L6 v2 for semantic embeddings tuned for retrieval quality.
- Built a stateful MemoryAgent that maintains conversational context across multiple turns and integrated it directly into the inference pipeline for coherent multi turn behaviour.
- Shipped the whole system behind a Streamlit interface as a deployed end to end tool, owning the design, implementation, and rollout in full.

CreditIQ, Fairness by Design Credit Scoring System

Python

scikit learn

AIF360

SHAP

Streamlit

- Lifted the model's Disparate Impact ratio from a failing 0.79 to a compliant 0.88, clearing the EU AI Act and AGG 80 percent fairness threshold.
- Diagnosed a hidden intersectional bias where younger women were still being penalised even after single axis age bias was fixed, using SHAP, then designed a four way age by gender threshold matrix to correct it without over correcting into reverse discrimination.
- Cut the false negative rate from 44 percent to 16.7 percent while holding accuracy at 75 percent, documenting the fairness accuracy trade off as a deliberate and regulator defensible decision.
- Shipped a Streamlit decision support tool giving finance managers a recommendation plus a plain language LLM generated explanation, keeping a human in the loop per GDPR Article 22 and EU AI Act Article 14.
- Backed the pipeline with unit tests at 100 percent branch coverage and a full regulatory write up covering EU AI Act Annex III, GDPR, a model card, and attack vectors.

Real Time Flight Tracking Data Pipeline

Python

PySpark

BigQuery

dbt

Apache Airflow

GCP Dataproc

GCE

GCS

Tableau

TabPy

OAuth2

- Set up real time data collection from the OpenSky Network API, polling live flight positions every 30 seconds and enriching each observation against airport, aircraft, and weather data from four external sources.
- Cleaned raw data with PySpark on Google Cloud and shaped it into analysis ready tables with dbt, computing each aircraft's nearest airport along the way and yielding more than 128 thousand joined records.
- Orchestrated the whole system with Apache Airflow so that the batch and real time layers refresh automatically every 15 minutes with no manual intervention.
- Surfaced findings in Tableau with Python statistics served through TabPy, showing that air traffic dropped by a factor of 4.4 in heavy rain and clustered heavily around hubs such as Frankfurt and Munich.

RESEARCH AND THESIS

Bachelor Thesis, Diabetes Prediction Using Machine Learning

GL Bajaj Institute of Technology and Management, IEEE style paper

- Built a full end to end machine learning pipeline comparing six classifiers on a clinical diabetes dataset of 768 patients, using 10 fold cross validation and per model confusion matrices.
- Caught biologically impossible zero values that the original authors had overlooked, then applied IQR based outlier removal and proper imputation to restore data integrity.
- Chose ROC AUC over accuracy as the headline metric for a 65 to 35 imbalanced dataset, avoiding a misleadingly rosy accuracy figure that would have hidden real error patterns.
- Wrote up findings as an IEEE style paper including an honest section on limitations and what to do differently in a follow up study.

LANGUAGES

English: fluent, professional working proficiency. **German:** B1, in progress.